

Acute and Chronic Inflammatory Dermatitis



Christina Kinnevey Greig, MD

(Slides adapted from Dr. Kim Pfothauer)

Objectives

◆ For EACH of the following disorders, know the objectives **A-E**

- ◆ 1. Atopic Dermatitis
- ◆ 2. Asteatotic (xerotic) Eczema
- ◆ 3. Nummular Eczema
- ◆ 4. Dyshidrotic Eczema
- ◆ 5. Contact Dermatitis (Irritant and allergic)
- ◆ 6. Seborrheic dermatitis
- ◆ 7. Stasis Dermatitis

◆ Objectives **A-E**

- ◆ A. Recognize the description, or a picture, of a typical clinical presentation
- ◆ B. Describe the most commonly affected areas
- ◆ C. Identify the key epidemiologic features
- ◆ D. List the etiology, triggers, and pathophysiology that initiate the acute and chronic inflammatory responses
- ◆ E. Choose the most appropriate therapy

Dermatitis Definition

◆ Dermatitis - Inflammation of the skin



Note: The words dermatitis and eczema are frequently used interchangeably although eczema typically refers to atopic dermatitis specifically

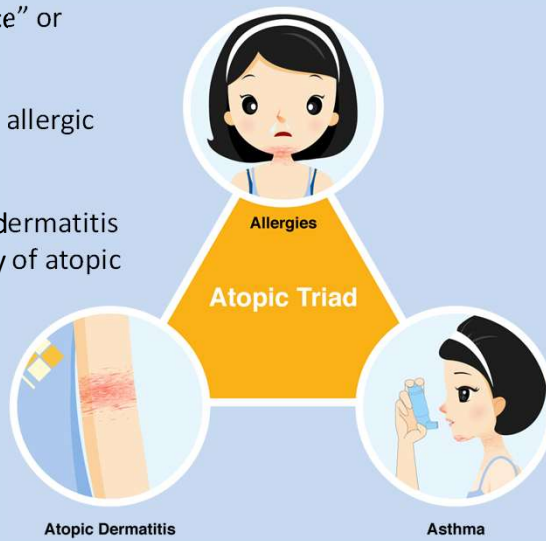
Atopic Dermatitis

- ◆ Chronic, pruritic inflammatory skin disease
- ◆ Also called “eczema” – from Greek – “to boil over, break out”
- ◆ Occurs most commonly in children but also affects many adults
 - ◆ Affects 10-20% of the US population
 - ◆ Typical onset is before age 5 to 7 years old (85%)
 - ◆ Presentation is common in the first year of life (60%)
- ◆ Prevalence has almost tripled in industrialized countries in the past 3 decades

prevalence is less in rural areas than in urban. Multiple theories on the reasons for increasing prevalence including importance of early childhood infections, early allergen exposure, early diet and early antibiotic use. Hygiene hypothesis.

Atopic Triad

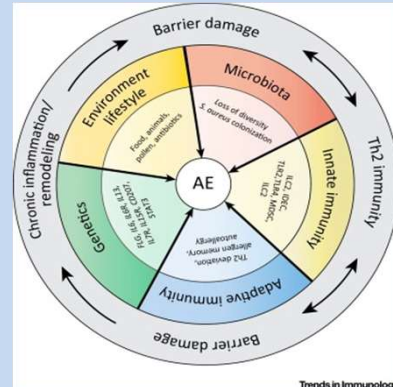
- ◆ Atopy: Greek for “out of place” or “unusual”
- ◆ Predisposition to developing allergic hypersensitivity reactions
- ◆ 70% of patients with atopic dermatitis have a positive family history of atopic diseases



Dermpatiented.com/atopicdermatitis/

Atopic Dermatitis Pathogenesis

- ◆ Appears to involve a genetic defect in the proteins supporting the epidermal barrier
- ◆ Often associated with elevated serum immunoglobulin E (IgE)
- ◆ Type I hypersensitivity
- ◆ Alterations in skin resident microbial flora
- ◆ Alterations in immunologic responses in T cells, antigen processing, inflammatory cytokine release, allergen sensitivity and infection
- ◆ Genetic basis influenced by environmental factors.



Trends in Immunology
[Image source](#)

Atopic Dermatitis Pruritis: “The Itch that Rashes”

- ❖ Constant scratching may lead to the destruction of melanocytes resulting in areas of **hypopigmentation** and **lichenification** (skin becomes thickened and leathery)



Lichenification



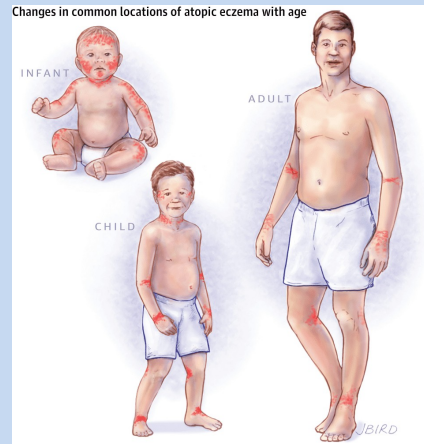
Post inflammatory Hypopigmentation



Atopic Dermatitis

extensor
surfaces and
face

- ◆ Infantile stage (birth to 2 years)
 - ◆ Dry, red scaling areas confined to cheeks but sparing the perioral and paranasal areas
 - ◆ Habitual lip licking by atopic child may lead to oozing, crusting and scaling on the lips and perioral skin
 - ◆ Spares diaper area
- ◆ Childhood (2-12 years)
 - ◆ Lichenified plaques in a flexural distribution
 - ◆ Especially of the antecubital and popliteal fosse, volar aspect of the wrists, ankles and neck
- ◆ Adult (>12 years)
 - ◆ More localized and lichenified
 - ◆ Similar distribution to childhood stage



Same area that becomes flushed when exposed to cold

Atopic dermatitis: infantile



Confluent erythema, microvesiculation, papules, crust, and scale of a young Asian infant; the shoulders are relatively spared, being protected from scratching by clothing.
Reproduced with permission from: Fitzpatrick, TB, Johnson, RA, Wolff, K et al (Eds). *Color Atlas and Synopsis of Clinical Dermatology*, 3rd ed, McGraw-Hill, New York, 1997. Copyright © McGraw-Hill.



Atopic dermatitis



limit hand washing and use hand sanitizer or wear protection from exposure to water

Atopic Hand Dermatitis

- ◆ Aggravated by wetting and washing with detergents, harsh soaps, and disinfectants
- ◆ Exfoliative dermatitis and erythroderma in patients with extensive skin involvement



Erythroderma is the term used to describe intense and usually widespread reddening of the skin due to inflammatory skin disease. It often precedes or is associated with exfoliation (skin peeling off in scales or layers), when it may also be known as exfoliative dermatitis

Atopic Dermatitis

- ◆ Diagnosis is clinical
 - ◆ Typical distribution sites and morphology of lesions
 - ◆ White dermatographism: stroking of involved skin will not lead to redness but to blanching
- ◆ DDx: Contact dermatitis, seborrheic dermatitis, psoriasis, drug reactions, scabies, Wiskott-Aldrich syndrome, hyperimmunoglobulin E syndrome, nutritional deficiencies, acrodermatitis enteropathica
- ◆ Blood work shows increased IgE in serum and eosinophilia

Atopic Dermatitis Complications

- ◆ Secondary infections
 - ◆ Bacterial (boils, folliculitis, impetigo)
 - ◆ Staph aureus most commonly
 - ◆ Herpes simplex virus (eczema herpeticum)



Atopic Dermatitis

- ◆ Untreated sites can persist for months to years
- ◆ Spontaneous remission in >40% of childhood cases
- ◆ 30-50% develop asthma and/or allergic rhinitis
- ◆ Can have life-threatening complications with severe secondary infections

Atopic Dermatitis: Treatment

- ◆ Stop the itch cycle!
 - ◆ Avoid scratching
 - ◆ Anti-pruritics
 - ◆ Topical and systemic such as antihistamines
- ◆ Topical Glucocorticoids
 - ◆ Limit use on face to limit atrophy
 - ◆ Very rarely need to use systemic steroids
- ◆ Topical calcineurin inhibitors (tacrolimus and pimecrolimus)
- ◆ UVA-UVB phototherapy
- ◆ Topical and/or systemic antibiotics/antivirals for secondary infections
- ◆ Limited use of soap
- ◆ Pat dry after shower
- ◆ Emollients!

Emollients

- ◆ No fragrances!
- ◆ Aim for creams over lotions
- ◆ Aquaphor, CeraVe, Cetaphil, Eucerin, etc.
- ◆ Vasoline
- ◆ Apply immediately after bathing



higher heat content washes off more oils

Asteatotic eczema

- ◆ Also called eczema craquele or xerotic eczema
- ◆ Clinically seen in the winter months
 - ◆ Commonly seen in the elderly
- ◆ Dry, scaly, skin resulting in erythema, cracks and fissuring on torso and extremities
 - ◆ Commonly on anterolateral legs
- ◆ Often called the “winter itch”
 - ◆ Pain often a symptom and is made worse by drying lotions such as calamine
- ◆ Treatment: topical steroids



Asteatotic (xerotic) eczema



[Photo source](#)



[Photo source](#)



[Photo source](#)

Various degrees of associated inflammation and erythema

Nummular dermatitis

- ◆ Also called nummular eczema or discoid eczema
- ◆ “Coin-shaped” (round-to-oval) lesion
- ◆ Most commonly found on the extremities and sometimes the trunk (head spared)
- ◆ Often start as erythematous, crusting patches which then coalesce into larger, dryer plaques with scaling and occasionally central clearing
- ◆ Common in the 6th to 7th decade of life (rare in children)
- ◆ DDX: tinea corporis, psoriasis
- ◆ Treatment: topical steroids, emollients



[Picture source](#)



[Picture source](#)



Nummular dermatitis

Dyshidrotic eczema

- ◆ Also called acute palmoplantar eczema or pompholyx
- ◆ Recurrent or chronic relapsing form of a very pruritic, vesicular eruption affecting palms, lateral and dorsal aspects of fingers or soles
- ◆ In 70-80% of people, only hands are involved
- ◆ Atopic dermatitis and exposure to irritants/allergens and stress are predisposing factors
- ◆ Episodes may recur at intervals of 3-4 weeks and more common in warm weather



Source: Klaus Wolff, Richard Allen Johnson, Arturo P. Saavedra:
Fitzpatrick's Color Atlas and Synopsis of Clinical Dermatology,
7th Edition, www.accessmedicine.com
Copyright © McGraw-Hill Education. All rights reserved.

Dyshidrotic eczema

- ◆ Mild form appears as red, scaly, weeping and pruritic vesicles which may rupture to form enlarging collarettes of scale
- ◆ Severe form occurs as large bullae on the palms and feet
- ◆ Most common in young adults
- ◆ Equally affects males and females
- ◆ Burning pain and intense pruritis





Dyshidrotic Eczema



- ◆ Diagnosis is clinical
- ◆ DDx: atopic hand dermatitis, contact dermatitis, tinea corporis (pedis), dermatophytid (id) reaction
- ◆ Mildly affected patients experience spontaneous resolution in 2-3 weeks
- ◆ Treatment: dry hands thoroughly after washing, remove jewelry before wet work, emollients, topical steroids, topical calcineurin inhibitors, systemic steroids for severe disease, phototherapy

assoc w/ fungal infection

A dermatophytid (id) reaction, or autoeczematization, is a secondary pruritic, papulovesicular eruption that may occur on the palms in association with a dermatophyte infection at a distant site (eg, on the scalp or foot) ([picture 8](#)). A complete skin examination and a KOH preparation of scrapings from suspicious

lesions are necessary to confirm the diagnosis.

Contact Dermatitis

Acute or chronic inflammatory reactions to substances that come in contact with the skin

◆ Irritant contact dermatitis (ICD) is

- ◆ caused by a chemical irritant
- ◆ Occurs in everyone (atopy increases risk)
- ◆ Dependent on concentration of offending agent
- ◆ Confined to the area of exposure and is therefore always sharply marginated and never spreads

◆ Allergic contact dermatitis (ACD)

- ◆ caused by an antigen that elicits a type IV (cell-mediated or delayed) hypersensitivity reaction
- ◆ Occurs only in sensitized individuals
- ◆ Immunologic reaction that tends to involve the surrounding skin (spreading phenomenon) and may even spread beyond affected sites
- ◆ Generalization may occur

Irritant Contact Dermatitis

- ◆ Localized disease confined to areas **exposed to irritants**
- ◆ Caused by exposure of skin to chemical and other physical agents that are capable of irritating the skin, acutely or chronically
- ◆ Severe irritants cause toxic reactions even after a short exposure
- ◆ Most cases are caused by chronic cumulative exposure to one or more irritants
- ◆ The hands are the most commonly affected area due to exposure
- ◆ Most common form of occupational skin disease



[Photo source](#)



Diaper dermatitis

[Photo source](#)

Note it spares the skin folds where urine does not come in contact (unlike candida diaper dermatitis)



[Photo source](#)

Evolution of lesions: erythema with a dull, non glistening surface -> vesiculation (or blister formation) -> erosion -> crusting -> shedding of crusts and scaling or erythema (chemical burns) -> necrosis -> shedding of necrotic tissue -> ulceration -> healing

Saliva is alkaline and can be irritating to the skin with prolonged exposure

Irritant Contact Dermatitis

Soaps

detergents
waterless
hand cleaners



Acids and Alkalis

hydrofluoric acid
cement
chromic acid
phosphorus
ethylene oxide
phenol
metal salts



Industrial solvents

coal tar solvents
petroleum
chlorinated
hydrocarbons
alcohol solvents
ethylene glycol ether
turpentine
ethyl ether
acetone
carbon dioxide
DMSO
dioxane
styrene

Irritant Contact Dermatitis

Plants

Euphorbiaceae (spurges,
crotons, poinsettias,
machneel tree)

Racunculaceae (buttercup)

Cruciferae (black mustard)

Urticaceae (nettles)

Solanaceae (pepper,
capsaicin)

Opuntia (prickly pear)



Other

fiberglass

wool

rough synthetic clothing

fire-retardant fabrics

"NCR" paper

Prolonged water contact

Irritant Contact Dermatitis

- ◆ May need patch test to r/o allergic contact dermatitis
- ◆ Healing usually occurs within 2 weeks of removal of noxious stimuli
- ◆ Prevention
 - ◆ Identify and remove the etiologic agent
 - ◆ Avoid irritant or caustic chemicals by wearing protective clothing (goggles, shields, gloves)
 - ◆ Barrier creams
 - ◆ If contact does occur, wash with water or weak neutralizing solution
- ◆ Topical or systemic glucocorticoids
- ◆ Lubricating/protective creams
- ◆ Oral or topical antibiotics for cracks and fissures to prevent/treat secondary infections

Allergic Contact Dermatitis

- ◆ Systemic disease defined by hapten-specific T cell mediated inflammation
- ◆ One of the most frequent, vexing, and costly skin problems
- ◆ Eczematous (papules, vesicles, pruruitc) dermatitis
- ◆ Due to **reexposure** to a substance to which the individual is sensitized
- ◆ Classic, delayed, cell-mediated hypersensitivity reaction

Allergic Contact Dermatitis

- ◆ Eruption starts in a sensitized individual **48 hrs to days after contact with allergen**
- ◆ Repeated exposures lead to a **crescendo reaction**
- ◆ Site of eruption is confined to the site of exposure
- ◆ Subjective symptoms are intense pruritus
- ◆ Constitutional symptoms: "Acute illness" syndrome, including fever, but only in severe cases

Top Ten Contact Allergens (North American Contact Dermatitis Group)

<i>Allergen</i>	<i>Principal Sources of Contact</i>
Nickel sulfate	Metals, metals in clothing, jewelry, catalyzing agents
Neomycin sulfate	Usually contained in creams, ointments
Balsam of Peru	Topical medications
Fragrance mix	Fragrances, cosmetics
Thimerosal	Antiseptics
Sodium gold thiosulfate	Medication
Formaldehyde	Disinfectant, curing agents, plastics
Quaternium-15	Disinfectant
Bacitracin	Ointments, powder
Cobalt chloride	Cement, galvanization, industrial oils, cooling agents, eyeshades

Allergic Contact Dermatitis



[Photo source](#)



Lipstick allergy

Source: Klaus Wolff, Richard Allen Johnson, Arturo P. Saavedra:
Fitzpatrick's Color Atlas and Synopsis of Clinical Dermatology,
7th Edition, www.accessmedicine.com
Copyright © McGraw-Hill Education. All rights reserved.



Paraphenylenediamine (PPD) in black hair dye



thiuram in rubber footwear



Latex allergy

<http://www.pcds.org.uk/clinical-guidance/eczema-contact-allergic-dermatitis-including-latex-and-rubber-allergy>



Nickel allergy from belt buckle

Allergic Contact Dermatitis Lesions

- ◆ Initially confined to area of contact with allergen
- ◆ Acute: Well-demarcated erythema and edema with nonumbilicated vesicles and/or papule
 - ◆ In severe reactions, can have bullae and confluent erosions exuding serum and crusts



Bullae

Allergic Contact Dermatitis Lesions

- ◆ Subacute: Plaques of mild erythema showing small, dry scales



Acute Contact Dermatitis Lesions

- ◆ Chronic: Plaques of lichenification, scaling with satellite, small, firm, rounded or flat-topped papule, excoriations, erythema and pigmentation



Allergic Phytodermatitis



- ◆ Specific type of allergic contact dermatitis
- ◆ Occurs in sensitized individuals after exposure to a wide variety of plant allergens
- ◆ Common is Rhus Dermatitis from poison ivy, poison oak, and poison sumac in the US
- ◆ Others in the same plant family: cashew trees, mango trees, Japanese lacquer trees, and ginkgo
- ◆ Allergy to an oleoresin called urushiol



Poison Ivy rash

Allergic Phytodermatitis



- ◆ Characterized by an acute, very pruritic, eczematous dermatitis, often in linear arrangement
- ◆ Lines created where plant draws across the skin
- ◆ Blisters do not contain the oleoresin and cannot spread the inflammation
- ◆ Diagnosis is clinical
- ◆ Typically resolves in 1-2 weeks
- ◆ Prevention: Washing with soap ASAP inactivates and removes surface oleoresin
 - ◆ After 10 min, only 50% can be removed, after 30 min only 10%, after 60 min, none
- ◆ Treatment: Wet compresses, topical steroids, prednisone (antihistamines not helpful)

Type IV hypersensitivity

Diffuse or unusual patterns may be from contaminated animal hair or clothing

Patch Testing

- ◆ Used in patients with contact dermatitis to identify offending substances (note: this is different than skin prick tests used to identify triggers of allergic rhinitis)
- ◆ Place suspected allergen on an unaffected site distant from the original eruption and leave in place for 48 hours
- ◆ Return 48 hours later to observe. Positive test shows erythema, papules or vesicles
- ◆ T.R.U.E. Test: Thin-Layer Rapid use Epicutaneous Test
- ◆ Tests for 36 allergens that are responsible for 80% of allergic contact dermatitis

Positive Patch Test Reaction



From left to right: +++ reaction; ++ reaction (note the microvesicles); + reaction; +/- reaction.

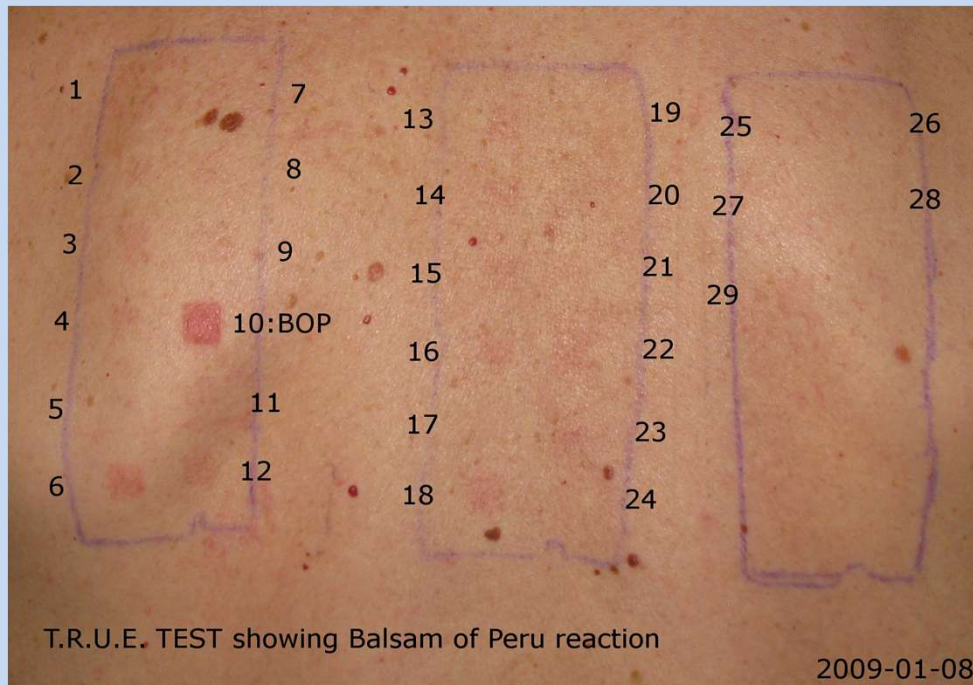
SkinSafe app, skinsafeproducts.com,
ewg.org/skindeep

Panel 1.3

1	☐	Nickel sulfate	7	☐	Colophony
2	☐	Wool Alcohols	8	☐	Paraben Mix
3	☐	Neomycin Sulfate	9	☐	Negative Control
4	☐	Potassium Dichromate	10	☐	Balsam of Peru
5	☐	Cane Mix	11	☐	Ethylendiamine Dihydrochloride
6	☐	Fragrance Mix	12	☐	Cobalt Dichloride

Millimeters





[Photo source](#)

Irritant vs. Allergic Contact Dermatitis

		Irritant CD	Allergic CD
Symptoms	Acute	Stinging, smarting →itching	Itching →pain
	Chronic	Itching/pain	Itching/pain
Lesions	Acute	Erythema → vesicles→erosions →crusts → scaling	Erythema → papules →vesicles→erosions → crust → scaling
	Chronic	Papules, plaques, fissures, scaling, crusts	Papules, plaques, scaling, crusts
Margination and site	Acute	Sharp, strictly confined to site of exposure	Sharp, confined to site of exposure but spreading in the periphery; usually tiny papules; may become generalized
	Chronic	Ill-defined	Ill-defined, spreads
Evolution	Acute	Rapid (few hours after exposure)	Not so rapid (12-72 h after exposure)
	Chronic	Months to years of repeated exposure	Months or longer; exacerbation after every reexposure
Causative agents		Dependent on concentration of agent and state of skin barrier; occurs only above threshold level	Relatively independent of amount applied, usually very low concentrations sufficient but depends on degree of sensitization
Incidence		May occur in practically everyone	Occurs only in the sensitized

[Chart source](#)

(FYI smarting is the sensation of a sharp, stinging pain)

Latex Allergy

- ◆ High risk groups include health care workers, rubber industry workers, and persons who have undergone multiple surgical procedures
- ◆ Three types of reactions:
 - ◆ Irritant contact dermatitis: non immune eczematous reaction caused by moisture, heat, and friction under gloves
 - ◆ Allergic contact dermatitis (Type IV): sensitization to rubber accelerators. Reaction limited to site of contact
 - ◆ Immediate-type hypersensitivity (Type I): IgE mediated reaction after previous sensitizations. Exposure in the air elicits allergic rhinitis, conjunctivitis, asthma, anaphylaxis and death
 - ◆ Often have a cross reaction with Bananas, Avocado, Chestnut, Kiwi, Passion fruit, maybe tomatoes (BLACK Passion)

Irritant: itching, erythema, and scaling
followed by thickened, crusted plaques.

use cotton liner under gloves

allergic: occurs in up to 10% of operating
room nurses. Accounts for about 80% of
cases of occupational acquired rubber
allergy

Immediate: patients very vulnerable in
hospital setting. anaphylaxis and death
can occur as a result of mucosal latex

absorption at the time of surgery

Seborrheic Dermatitis

- ◆ Very common skin condition characterized by redness and scaling
- ◆ Occurs in regions where sebaceous glands are most active
 - ◆ Face and scalp, the presternal area, and in body folds
- ◆ Mild scalp SD causes flaking (dandruff)
 - ◆ AKA: Cradle cap (infants), pityriasis sicca (dandruff)
- ◆ *Malassezia furfur* fungus is thought to play a role in the pathogenesis





Seborrheic Dermatitis

- ◆ Age of onset:
 - ◆ Infancy (within the first months)
 - ◆ Puberty
 - ◆ Most between 20-50 years or older
- ◆ More common in males
- ◆ Prevalence
 - ◆ 1-3% of the adult population
 - ◆ Increased in patients with HIV or Parkinson's disease



more immunocompromised and susceptible to infections

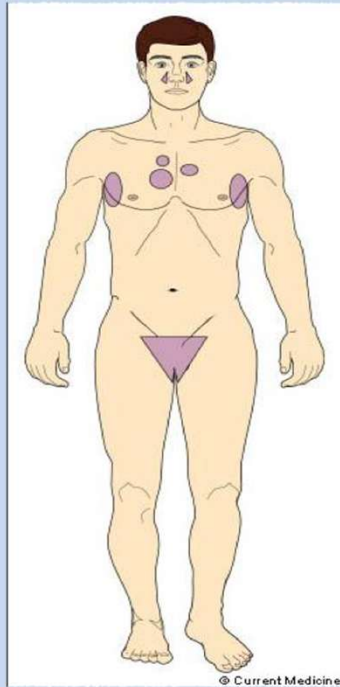
<https://www.hiv.uw.edu/go/basic-primary-care/cutaneous-manifestations/core-concept/all>

Seborrheic Dermatitis

◆ Lesions:

- ◆ Orange-red or gray-white skin
 - ◆ Often with “greasy” or white dry scaling macules, papules of varying size (5-20mm), or patches
 - ◆ Sharply margined
- ◆ Behind external ear: sticky crusts and fissures
- ◆ Scalp: diffuse, marked scaling (dandruff)
- ◆ Face & Trunk: scattered, discrete lesions
- ◆ Nummular, polycyclic, and even annular on the trunk





Source: Klaus Wolff, Richard Allen Johnson, Arturo P. Saavedra:
Fitzpatrick's Color Atlas and Synopsis of Clinical Dermatology,
7th Edition, www.accessmedicine.com
Copyright © McGraw-Hill Education. All rights reserved.

oil type glands

Seborrheic Dermatitis Treatment

- ◆ Behavior modification (stop scratching)
- ◆ Early treatment of flares
- ◆ Topical corticosteroids (low potency)
- ◆ Topical antifungals (ketoconazole)
- ◆ Alternatives include calcineurin inhibitors, sulfur or sulfonamide combinations
- ◆ If severe: systemic antifungals or oral retinoids
- ◆ Cradle cap responds to application of emollients like mineral oil to loosen the scales and brushing with soft brush
- ◆ Shampoos containing ketoconazole or anti proliferative agents: salicylic acid, selenium, tar, sulfur or zinc



Wolff K, Johnson RA, Suromond, D: Fitzpatrick's Color Atlas & Synopsis of Clinical Dermatology, 5th Edition: <http://www.accessmedicine.com>. Copyright © The McGraw-Hill Companies, Inc. All rights reserved.



Wolff K, Johnson RA, Suromond, D: Fitzpatrick's Color Atlas & Synopsis of Clinical Dermatology, 5th Edition: <http://www.accessmedicine.com>. Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Application of an emollient (white petrolatum, vegetable oil, mineral oil, baby oil) to the scalp (overnight, if necessary) to loosen the scales, followed by removal of scales with a soft brush (eg, a soft toothbrush) or fine-tooth comb

Stasis Dermatitis

- ❖ Common in lower extremities of patients with chronic venous insufficiency
- ❖ Affects 6.2% of patients over the age of 65
- ❖ Typically presents with erythematous, scaling, and eczematous patches or plaques on chronically edematous legs
- ❖ Acute forms may present with severely inflamed, weeping plaques, vesiculation, and crusting
- ❖ Chronic forms are characterized by hyperpigmentation, due to dermal hemosiderin deposition, scaling, and potential development of lipodermatosclerosis (fat degeneration and fibrosis)



Stasis Dermatitis



Acute weeping ulceration



Residual hyperpigmentation from hemosiderin deposition

Stasis Dermatitis Treatment

- ◆ Management involves the treatment of the underlying chronic venous insufficiency
 - ◆ Leg elevation
 - ◆ Weight reduction
 - ◆ Compression therapy
 - ◆ Venous ablation therapy
- ◆ Symptomatic treatment of skin dryness, pruritus and prevention of ulceration
- ◆ Topical corticosteroids for acute erythema, pruritus, vesiculation, and oozing



Further Reading

- ◆ American Academy of Dermatology
 - ◆ Student Modules
- ◆ Fitzpatrick's Color Atlas & Synopsis of Clinical Dermatology
- ◆ Habif: Clinical Dermatology
- ◆ DermIS.net

References

Basics: Fitzpatrick's Color Atlas & Synopsis of Clinical Dermatology, 7e (AccessMedicine)

Part I, Section 3 Psoriasis and Psoriasiform Dermatoses (Psoriasis, pityriasis rubra pilaris, pityriasis rosea)

Part I, Sections 32-34 and Part II, Section 14 (Lichen Planus)

Details: Fitzpatrick's Dermatology in General Medicine, 8e (AccessMedicine)

Section 4. Inflammatory Disorders Based on T-Cell Reactivity and Dysregulation (psoriasis, pityriasis rubra pilaris, lichen planus)

Section 6. Inflammatory Diseases Based on Abnormal Humoral Reactivity and Other Inflammatory Diseases (pityriasis rosea)

DermIS <http://www.dermis.net/dermisroot/en/home/index.htm>

Habif TP. Clinical Dermatology, 5th ed. Elsevier, 2009. (Available online via Touro's electronic library link with ClinicalKey)

UpToDate